

Name..... Adm. No..... Branch..... Sec.....

1. A ball is drawn from an urn containing 3 white and 3 black balls. After the ball is drawn, it is replaced and another ball is drawn. This process goes on indefinitely. What is the probability that, of the first 4 balls drawn, exactly 2 are white?

A) $\frac{1}{2}$ B) $\frac{1}{4}$ ☒ C) $\frac{3}{8}$ D) $\frac{3}{5}$

mark: 1

2. You go to a party with 500 guests. What is the probability that exactly one other guest has the same birthday as you? Calculate this approximately by using the Poisson pmf. (For simplicity, exclude birthdays on February 29.)

☒ A) 0.348 B) 0.455 C) 0.207 D) 0.856

mark: 1

3. Which distribution is a special case of Negative Binomial (NB(r, p)) distribution with r=1?

A) Binomial ☒ B) Geometric C) Poisson D) Hyper-Geometric distribution

mark: 1

4. The amount of time, in minutes, that a person must wait for a bus is uniformly distributed between zero and 15 minutes, inclusive.

i). On the average, how long must a person wait?

ii). Ninety percent of the time, the time a person must wait falls below what value?

☒ A) i) 7.5 minutes ii) 13.5 minutes B) i) 7.5 minutes ii) 1.5 minutes C) i) 10 minutes ii) 2.5 minutes D) i) 15 minutes ii) 13.5 minutes

mark: 1+1

5. The mean of a normal distribution is 50 and 5% of the values are greater than 60. Find the standard deviation of the distribution. [Given, $P(0 \leq z \leq 1.64) = 0.45$, $P(0 \leq z \leq 1.28) = 0.40$ where $z \sim N(0, 1)$]

☒ A) 6.1 B) 8.3 C) 5.5 D) 11.2

mark: 1

6. A lawyer commutes daily from his suburban home to his midtown office. The average time for a one-way trip is 24 minutes, with a standard deviation of 3.8 minutes. Assume the distribution of trip times to be normally distributed. If the office opens at 9:00 A.M. and the lawyer leaves his house at 8:40 A.M. daily, what percentage of the time is he late for work? [Given $P(0 \leq z \leq 1.053) = 0.354$, $P(0 \leq z \leq 1.310) = 0.405$, $P(-1.053 \leq z \leq 0) = 0.146$ where $z \sim N(0, 1)$]

A) 99.1 % ☒ B) 85.4 % C) 14.6% D) 90.5%

mark: 1

7. Suppose that a system contains a certain type of component whose time (in years) to failure is given by T modeled by the exponential distribution with mean time to failure $\beta = 5$. If 5 of these components are installed in different systems, what is the probability that at least 2 are still functioning at the end of 8 years?

A) 0.6582 B) 0.3557 ☒ C) 0.2627 D) 0.9933

mark: 2

8. A study involving stress is done on a college campus among the students. The stress scores follow a uniform distribution with the lowest stress score equal to 1 and the highest equal to 5. Using a sample of 75 students, find the 90th percentile for the average of 75 stress scores applying CLT. [Given, $P(0 \leq z \leq 1.64) = 0.45$, $P(0 \leq z \leq 1.28) = 0.40$ where $z \sim N(0, 1)$]

A) 2.8 ☒ B) 3.2 C) 3.7 D) 5.6

mark: 1